Problem: Machine Learning Based Regression Task (25 Marks):

The "House_Price_Prediction" dataset is designed to represent properties with features like size, number of bedrooms, number of bathrooms, age, and proximity to the city center. The target variable is the property price. Given the dataset, perform the following tasks to predict the price of properties based on various features.

a.) Data Preprocessing (8 marks)

- Handle any missing values present in the dataset. Describe the method you chose and justify your choice. (5 marks)
- Normalize the features of the dataset. (2 marks)
- Split the data into training, and test sets. (1 marks)

b.) Model Building (10 marks)

- Use Multi Linear Regression to develop a prediction model. (5 marks)
- Train the model using the training set and test it using the test set. (2marks)
- Evaluate the model's performance on the test set and report the mean squared error, R2. Plot this. Is the model overfitting or underfitting or aptly fitting? (3 marks)

c.) Analysis (7 marks)

- Visualize the distribution of actual vs. predicted property prices. (3 marks)
- Discuss any patterns or insights you can derive from the model's predictions. (2 marks)
- Provide any recommendations for improving the model's performance. (2 marks)